

Self-Reflection: Formative One Assignment

Takeout

First, I have to admit that having prior lessons and knowledge of programming, especially in Scratch, Python, and Py games, helped me so much in building and accomplishing this Grade Tracker assignment. But it was also a great challenge to understand how to solve real-life problems with programming and sharpen my thinking skills.

The most fascinating thing is that I improved my skills in using Object-Oriented Programming (OOP) in creating different classes for Assignments, Homework, and Exams. This made the coding easier and shorter; I think this was the best thing about the assignment. It was also associated with what we have covered for the past seven weeks in the programming one class.

Another important take from the assignment is that I learned the importance of modularity. This is simply breaking an extensive block of code into simpler blocks that are easy to follow, debug, and execute. For my modularity, the code was broken into three separate, organized files (main.py, assignments.py, and validation.py); this showed me how real developers keep their projects clean and readable.

Finally, writing the validation part taught me a very important lesson. This code is as good as how the end user utilizes it with minimum errors. I learned how to protect my program from crashing when a user accidentally input that is invalid for any reason. This was like putting safe guard to make sure the program runs smoothly.

Challenges I Faced

Coding was challenging, but for me the absolute hardest and most confusing part of this assignment was using Git and associated commands. I took many hours to figure this out, leaving out the YouTube sessions. The process of cloning the repository, linking histories, and pushing my code to GitHub was very overwhelming.

When I tried to pull from the repository that was online to the local repository that was cloned from the facilitator account, I faced many errors. Also, creating divergent branches was challenging. This happened because the history on my local machine and the GitHub repository did not match up. I had to research the error and manually type `git config pull.rebase false` just to fix the merging rules before I could git push, which was also hard because at first I did not know git commit, git add, and other associated commands. This was also because I am using Linux on my computer.

Even after all that, keeping track of staging changes was tricky. When I was updating my documentation, checking my git status showed me that I had a newly created Readme.md file waiting to be committed, while an older README.md file was marked as deleted. It was a lot to keep track of! Thankfully, I did eventually manage to get everything synced and successfully pushed my Main.py file in my first commit.

How I Intend to Improve It Given More Time

If I had more time to chose on what to improve on this assignment I would focus on two areas First is that the program is only a short time storage program (single use) as soon as the program is terminated the user input is deleted and the session is ended. This is heartbreaking for me remembering all the coding and git put into the application. Therefore I would like to add a simple JSON file on the computer to save the user input for other session once the program is terminated.

Also, I would like to increase the hours on git and github just creating dummy projects to practice my Git commands and interference from both the bash terminal and visual code studio. I want the process of adding, removing, branching, committing, and pushing to be more easier so it does not slow down my actual coding in the future assignment and lessons.